



Quadratic equations: what we can already solve

HANDOUT

GOAL

In this lesson

Write any quadratic equation in the form $ax^2 + bx + c = 0$, and solve those that are a product equal to zero or a square equal to a number.

BY THE END OF THE LESSON

- ☐ I can write a quadratic equation in the form $ax^2 + bx + c = 0$, name a, b and c with their signs, and tell a complete equation from an incomplete one.
- ☐ I can solve an equation whose left side is a product of linear factors and whose right side is 0, and an equation of the form $(x + p)^2 = d$.

WHAT YOU NEED FROM BEFORE

- ↔ Linear equations and forming an equation
- ↔ Square roots
- ↔ Expanding brackets and the square formulas
- ↔ Incomplete quadratic equations

1 class period • about 45 min

HOOK

Five equations

Here are five quadratic equations:

- (A) $3x^2 = 27$
- (B) $x^2 + 4x = 0$
- (C) $(x - 3)(x + 5) = 0$
- (D) $(x - 2)^2 = 9$
- (E) $x^2 - 4x - 5 = 0$

(A) and (B) are grade 8 equations, like the ones you just solved. (C) and (D) look different. Try (D) before reading on.

Level 1 • threshold

Solve (D): $(x - 2)^2 = 9$.

(E) looks like the hardest of the five. Is it?

Which equations are easy, and why

One equation, two faces

Multiply out the left side of (D):

$$\begin{array}{l|l} (x-2)^2 = 9 & \text{expand } (x-2)^2 \\ x^2 - 4x + 4 = 9 & -9 \\ x^2 - 4x - 5 = 0 & \end{array}$$

That is (E). So (E) and (D) are the **same equation**, and (E) has the same roots, 5 and -1 . Check: $5^2 - 4 \cdot 5 - 5 = 0$ and $(-1)^2 - 4 \cdot (-1) - 5 = 0$.

How easy an equation is depends on **how it is written**. Written as (D), it takes one line.

The general form

An equation that can be written as

$$ax^2 + bx + c = 0, \quad a \neq 0,$$

is a **quadratic equation**. The numbers a , b and c are its **coefficients**: a is the coefficient of x^2 , b of x , and c is the **constant term**.

To read them, first move every term to the left side and collect like terms. Then read each coefficient **with its sign**:

- (E) $x^2 - 4x - 5 = 0$: $a = 1$, $b = -4$, $c = -5$.
- (A) $3x^2 = 27$ becomes $3x^2 - 27 = 0$: $a = 3$, $b = 0$, $c = -27$.
- (B) $x^2 + 4x = 0$: $a = 1$, $b = 4$, $c = 0$.

Why $a \neq 0$? If $a = 0$, the x^2 term disappears and the equation is not quadratic any more (it is linear if $b \neq 0$).

Complete, incomplete, or not quadratic at all

A quadratic equation is **incomplete** if $b = 0$ or $c = 0$, like (A) and (B). You solved these in grade 8. If $b \neq 0$ and $c \neq 0$, it is **complete**, like (E).

Decide only **after** tidying. Some equations only look quadratic:

$$\begin{array}{l|l} (x+1)^2 = x^2 + 3 & \text{expand} \\ x^2 + 2x + 1 = x^2 + 3 & -x^2 - 3 \\ 2x - 2 = 0 & \end{array}$$

The x^2 cancels, so this equation is **linear** ($x = 1$).

Also not quadratic as written: $\frac{4}{x^2} - 1 = 0$ (x is in a denominator, so $x \neq 0$ is needed; as written, it is not of the form $ax^2 + bx + c = 0$), and $3^2x + x - 4 = 0$ ($3^2 = 9$ is just a number, so this is $10x - 4 = 0$).

A product equal to zero

Zero-product rule. A product is equal to 0 if and only if at least one of its factors is 0:

$$p \cdot q = 0 \iff p = 0 \text{ or } q = 0.$$

Why: if $p \neq 0$, divide both sides by p and get $q = 0$.

So (C) splits into two linear equations:

$(x - 3)(x + 5) = 0$	zero-product rule
$x - 3 = 0 \text{ or } x + 5 = 0$	solve each
$x = 3 \text{ or } x = -5$	

Grade 8's $x^2 + 4x = 0$ uses the same rule: $x(x + 4) = 0$, so $x = 0$ or $x = -4$.

Only zero works like this. From $p \cdot q = 6$ you cannot find p : $1 \cdot 6$, $2 \cdot 3$ and $\frac{1}{2} \cdot 12$ all give 6. You only learn that $p \neq 0$.

A square equal to a number

In $(x + p)^2 = d$, the bracket $x + p$ is a number whose square is d . There are three cases:

- $d > 0$, e.g. $(x - 2)^2 = 9$: $x - 2 = 3$ or $x - 2 = -3$. **Two roots**, $x = 5$ and $x = -1$.
- $d = 0$, e.g. $(x - 2)^2 = 0$: $x - 2 = 0$. **One root**, $x = 2$.
- $d < 0$, e.g. $(x - 2)^2 = -1$: no real number has a negative square. **No real roots**.

We always look for roots among the real numbers. That is why the third case is called **no real roots**.

In general, for $d > 0$: $x + p = \sqrt{d}$ or $x + p = -\sqrt{d}$. Write both equations out, then solve each one. That avoids sign slips.

What we can solve so far

- **Incomplete**, $ax^2 + bx = 0$, like $x^2 + 4x = 0$: write $x(ax + b) = 0$ and use the zero-product rule.
- **Incomplete**, $ax^2 + c = 0$, like $3x^2 = 27$: bring it to $x^2 = d$.
- **A product of linear factors equal to 0**, like $(x - 3)(x + 5) = 0$: the zero-product rule.
- **A square equal to a number**, like $(x - 2)^2 = 9$: $x + p = \sqrt{d}$ or $x + p = -\sqrt{d}$.
- **Other equations**, like $x^2 - 4x - 5 = 0$: we have no general way yet to choose a useful form.

(E) was easy because we already knew its form (D).

Worked examples

Example 1 • The general form of $5x - 2x^2 = 3$

MOVE EVERYTHING TO THE LEFT

$5x - 2x^2 - 3 = 0$, that is, $-2x^2 + 5x - 3 = 0$.

MAKE A POSITIVE

Multiply both sides by -1 : $2x^2 - 5x + 3 = 0$. The roots do not change.

COEFFICIENTS

$a = 2$, $b = -5$, $c = 3$. (From $-2x^2 + 5x - 3 = 0$, the answer $a = -2$, $b = 5$, $c = -3$ is also correct: it is the same equation.)

COMPLETE OR INCOMPLETE

$b \neq 0$ and $c \neq 0$, so the equation is complete.

Example 2 • A product equal to zero: $(x - 4)(2x + 3) = 0$

ZERO-PRODUCT RULE

$x - 4 = 0$ or $2x + 3 = 0$.

SOLVE EACH

$x = 4$ or $2x = -3$, so $x = -\frac{3}{2}$.

CHECK

$x = 4$: $0 \cdot 11 = 0 \checkmark$. $x = -\frac{3}{2}$: $(-\frac{11}{2}) \cdot 0 = 0 \checkmark$.

Example 3 • A square equal to a number

$$(x + 2)^2 = 16$$

$16 > 0$: $x + 2 = 4$ or $x + 2 = -4$, so $x = 2$ or $x = -6$.

$$(x + 4)^2 = 0$$

$x + 4 = 0$, so there is one root, $x = -4$.

$$(x + 3)^2 = -4$$

A square cannot be negative, so the equation has no real roots.

SUMMARY

What to remember

- A **quadratic equation** can be written as $ax^2 + bx + c = 0$ with $a \neq 0$. Move every term to one side first, then read a, b, c **with their signs**.
- It is **incomplete** if $b = 0$ or $c = 0$, and **complete** otherwise. Decide after tidying: sometimes the x^2 cancels.
- **Zero-product rule:** $p \cdot q = 0$ exactly when $p = 0$ or $q = 0$. It works only with 0 on the other side.
- **A square equal to a number, $(x + p)^2 = d$:**
 - $d > 0$: $x + p = \sqrt{d}$ or $x + p = -\sqrt{d}$, so there are two roots;
 - $d = 0$: one root, $x = -p$;
 - $d < 0$: no real roots.
- Check a root by substituting it into the original equation.

NEXT LESSON

$x^2 - 4x - 5 = 0$ turned out to be $(x - 2)^2 = 9$ in disguise. Can every quadratic equation be rewritten as a square equal to a number?

Lesson 1.2 · Solving by completing the square